# eventPublish

## Service Description

Service ID: *"eventPublish"*

**Abstract**
This document describes the `eventPublish` service, version 1.0.0, produced by the EventHandler support system of Arrowhead 5.2.2.

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# 1 Overview

This document describes the `eventPublish` service, which provides publication of events to the subscribers of their event type. It is produced by the EventHandler system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

## 1.1 Significant Prior Art

Successor of the Arrowhead 4.6 EventHandler publish service.

## 1.2 How This Service Is Meant to Be Used

A publisher calls `publish` with the event type, payload and metadata; the EventHandler posts the event to every matching subscriber's `notifyUri` with its own certificate and identity token.

## 1.3 Important Delimitations and Dependencies

Delivery is asynchronous and best effort; the result only counts matched subscriptions and accepted deliveries. With `service.authorization.enabled=true` publishing needs a grant.

# 2 Service Interface

This section describes the interfaces to the `eventPublish` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

Figure 1: SysML-style block of the `eventPublish` service and its operations.

The following interface operations are available.

## 2.1 operation publish (EventPublishRequest) : EventPublishResult

Publishes an event to the matching subscriptions.

# 3 Information Model

Here, all data objects that can be part of the `eventPublish` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.3, which are used to represent things like hashes and identifiers.

## 3.1 struct EventPublishRequest

An event to publish.

| Field | Type | Description |
|---|---|---|
| eventType | String | Event type, camelCase. |
| payload | String | Event payload (text). |
| metadata | Metadata | Event metadata, matched against subscription filters. |
| timestamp | String | Time of the event (RFC 3339). |

## 3.2 struct EventPublishResult

Outcome of a publication.

| Field | Type | Description |
|---|---|---|
| matchedSubscriptions | Number | Number of subscriptions the event matched. |
| acceptedDeliveries | Number | Number of notifications accepted for delivery. |

## 3.3 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

| Type | Description |
|---|---|
| Metadata | A set of key-value pairs with arbitrary, possibly nested values. |
| Number | An integer number. |
| String | A chain of characters. |

# 4 References

[1] J. Delsing, “EventHandler – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/EventHandler/documentation/v5.2.2/EventHandler-v5.2.2-SysD.pdf.

[2] ——, “eventPublish HTTP/TLS/JSON – Interface Design Description (EventHandler),” Eclipse Arrowhead, IDD 5.2.2, 2026, eventHandler-v5.2.2-eventPublish-HTTP-IDD.pdf.

# 5 Revision History

## 5.1 Amendments

| No. | Date | Version | Subject of Amendments | Author |
|---|---|---|---|---|
| 1 | 2026-10-22 | 5.2.2 | Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace. | Jerker Delsing |
| 2 | 2026-10-30 | 5.2.2 | Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged. | Claude Opus 5.5 (draft) |

## 5.2 Quality Assurance

| No. | Date | Version | Approved by |
|---|---|---|---|
| 1 | | 5.2.2 | |